

CRYPTOGRAPHY AND NETWORK SECURITY

Mathematical background notes

1. MODULAR ARITHMETIC

Modular arithmetic is arithmetic based on the remainder after division.

Example:

$$17 \bmod 5 = 2$$

$$\text{Because: } 17 = 5 \times 3 + 2$$

Therefore, the remainder is 2.

2. CONGRUENCE

Congruence means that two numbers have the same remainder when divided by a number.

It is written as:

$$a \equiv b \pmod{n}$$

Example:

$$17 \equiv 5 \pmod{12}$$

$$(7 \bmod 12 = 5 \text{ and}$$

$$5 \bmod 12 = 5)$$

Both have the same remainder.

Therefore:

$$17 \equiv 5 \pmod{12}$$

3. GROUPS

A group is a set of elements with an operation that follows four important rules.

The four rules are:

1. Closure
2. Associativity
3. Identity
4. Inverse

1. Closure:

If we perform an operation on two elements of the group, the result must also belong to the group.

Set: $\{1,2,3,4,5\}$

Example:

$$2 + 3 = 5$$

If 2, 3 and 5 are all in the set, closure is satisfied.

2. Associativity:

$$(a + b) + c = a + (b + c)$$

Example:

$$(2 + 3) + 4 = 2 + (3 + 4)$$

Left side: $5 + 4 = 9$

Right side: $2 + 7 = 9$

$$\text{L.H.S.} = \text{R.H.S.}$$

3. Identity:

Identity is a value that does not change another value.

For addition:

$$a + 0 = a$$

Therefore, 0 is the identity for addition.

For multiplication:

$$a \times 1 = a$$

Therefore, 1 is the identity for multiplication.

4. Inverse:

An inverse is a value that gives the identity when combined with another value.

For addition:

$$5 + (-5) = 0$$

Therefore, -5 is the additive inverse of 5.

For multiplication:

$$5 \times 1/5 = 1$$

Therefore, $1/5$ is the multiplicative inverse of 5.

Therefore, **GROUP = Closure + Associativity + Identity + Inverse**

4. RINGS

A ring is a mathematical system in which addition and multiplication can be performed.

For example, integers form a ring.

Integers are:

..., -3, -2, -1, 0, 1, 2, 3, ...

Example:

$$2 + 3 = 5$$

$$2 \times 3 = 6$$

The answers are also integers.

Therefore, integers form a ring under normal addition and multiplication.

5. FINITE FIELDS

A **Galois field** (also called a **finite field**) is a mathematical set with a fixed, limited number of elements in which you can add, subtract, multiply, and divide without ever leaving the set.

A finite field is usually written as:

$$\text{GF}(n)$$

Example:

$$\text{GF}(5) = \{0, 1, 2, 3, 4\}$$

There are only 5 elements.

Calculations are performed modulo 5.

Example:

$$3 + 4 = 7$$

$$7 \bmod 5 = 2$$

Therefore:

$$3 + 4 = 2 \text{ in } \text{GF}(5)$$

Finite fields are important in cryptography. They are used in cryptographic algorithms such as AES.

6. LINEAR CONGRUENCE

A linear congruence is an equation of the form:

$$ax \equiv b \pmod{n}$$

The main purpose is to find x .

Important condition: The equation $ax \equiv b \pmod{n}$ has a solution if:

$$\gcd(a, n) \text{ divides } b.$$

Example:

$$3x \equiv 6 \pmod{7}$$

Try $x = 2$:

$$3 \times 2 = 6$$

Therefore:

$$6 \equiv 6 \pmod{7}$$

So: $x = 2$

Another example:

$$2x \equiv 4 \pmod{6}$$

Try $x = 2$:

$$2 \times 2 = 4$$

Therefore: $x = 2$

7. MODULAR INVERSE

(Already discussed in class) The modular inverse of a number is a number that gives 1 when multiplied with the original number under modulo.

We want:

$$ax \equiv 1 \pmod{n}$$

Example:

Find the inverse of 3 modulo 7.

We need: $3x \equiv 1 \pmod{7}$

Solution Try yourself

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the modular inverse of 3 modulo 7 is 5.

8. EULER'S TOTIENT FUNCTION

Euler's Totient Function is written as:

$$\phi(n)$$

It tells us how many numbers from 1 to n are relatively prime to n.

Two numbers are relatively prime if their GCD is 1.

Example:

Find $\phi(8)$.

Numbers from 1 to 8 are:

1, 2, 3, 4, 5, 6, 7, 8

Numbers relatively prime to 8 are:

1, 3, 5, 7

There are 4 numbers.

Therefore:

$$\phi(8) = 4$$

Important formula:

- If p is a prime number:
 $\phi(p) = p - 1$

Example:

$$\phi(7) = 7 - 1 = 6$$

Another important formula:

If p and q are different prime numbers:

$$\phi(pq) = (p - 1)(q - 1)$$

Example: $\phi(15)$

$$p = 3$$

$$q = 5$$

$$\phi(15) = (3 - 1)(5 - 1)$$

$$\phi(15) = 2 \times 4$$

$$\phi(15) = 8$$

9. EULER'S THEOREM

Euler's theorem states:

If $\gcd(a,n) = 1$, then:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

If a and n are relatively prime, then a raised to $\phi(n)$ gives remainder 1 when divided by n .

Example:

$$a = 3$$

$$n = 10$$

First:

$$\gcd(3,10) = 1$$

Now:

$$\phi(10) = 4$$

Therefore:

$$3^4 \equiv 1 \pmod{10}$$

Verify:

$$3^4 = 81$$

$$81 \bmod 10 = 1$$

Therefore:

$$3^4 \equiv 1 \pmod{10}$$

10. FERMAT'S LITTLE THEOREM

Fermat's Little Theorem is mainly used when the modulus is a prime number.

$$a^{(p-1)} \equiv 1 \pmod{p}$$

where p is a prime number and p does not divide a .

Example:

$$a = 2$$

$$p = 7$$

$$\text{Then: } 2^{(7-1)} = 2^6$$

$$2^6 = 64$$

$$64 \bmod 7 = 1$$

Therefore:

$$2^6 \equiv 1 \pmod{7}$$

- *Difference:*

$$\text{Euler's Theorem: } a^{\phi(n)} \equiv 1 \pmod{n}$$

Fermat's Little Theorem: $a^{(p-1)} \equiv 1 \pmod{p}$ (Fermat's theorem is used when the modulus is prime.)

11. PRIMALITY TESTING

A prime number has exactly two positive factors:

1 and itself.

Examples:

2, 3, 5, 7, 11, 13, 17,

Primality testing means checking whether a number is prime or not.

Example:

Check whether 17 is prime.

We try dividing 17 by smaller numbers.

17 is not divisible by 2, 3, 4

Therefore:

17 is prime.

For larger numbers, we can check divisibility only up to $\sqrt{n}$.

Cryptography uses advanced primality tests such as:

1. Fermat Primality Test
2. Miller-Rabin Primality Test

Fermat Test: Calculate: $a^{(n-1)} \bmod n$

Result $\neq 1 \rightarrow$ Definitely composite

Result = 1 $\rightarrow$ Probably prime

EXAMPLE 1:

Check whether 7 is prime using Fermat's test.

$$n = 7$$

Choose:

$$a = 2 \quad (1 < a < n)$$

Calculate:

$$2^{(7-1)} \bmod 7$$

$$= 2^6 \bmod 7$$

$$= 64 \bmod 7$$

$$= 1$$

The result is 1.

Therefore:

7 is probably prime.

EXAMPLE 2: Check whether 8 is prime.

$$n = 8$$

Choose: $a = 3$

Calculate:

$$3^{(8-1)} \bmod 8$$

$$= 3^7 \bmod 8$$

We only need the remainder.

$$3^2 = 9 \equiv 1 \pmod{8}$$

Therefore:

$$3^6 \equiv 1 \pmod{8} \quad \{3^6 \equiv 1 \times 1 \times 1 \pmod{8}\}$$

$$3^7 \equiv 3 \pmod{8} \quad \{3^7 = 3^6 \times 3\}$$

The result is 3, not 1.

Therefore:

8 is definitely NOT prime.

Miller–Rabin Test:

1. Calculate .
2. Choose a base .
3. Calculate .
4. If or , **PASS**.
5. Otherwise, square repeatedly.
6. If you get , **PASS**.
7. If you never get , **COMPOSITE**.

Example: Check whether 13 is a prime number or not.

Step 1: Calculate $n - 1$

$$n = 13$$

$$n - 1 = 13 - 1 = 12$$

Write 12 in the form:

$$n - 1 = 2^s \times d$$

$$12 = 2^2 \times 3$$

Therefore:

$$s = 2$$

$$d = 3$$

Step 2: Choose a base

$$\text{Let } a = 2$$

Step 3: Calculate $a^d \bmod n$

$$x = 2^3 \bmod 13$$

$$x = 8$$

Step 4: Check the result

We check whether x is:

$$1 \text{ OR } n - 1$$

Here:

$$x = 8$$

$$n - 1 = 12$$

Since 8 is neither 1 nor 12, we continue.

Step 5: Square the result

$$x = 8^2 \bmod 13$$

$$x = 64 \bmod 13$$

$$x = 12$$

Now:

$$x = n - 1$$

because:

$$12 = 13 - 1$$

Therefore, 13 passes the Miller-Rabin test.

Therefore, 13 is PROBABLY PRIME.

12. PRIME FACTORIZATION

Prime factorization means breaking a number into its prime factors.

Example:

Find the prime factorization of 60.

Start with 2:

$$60 = 2 \times 30$$

Again divide by 2:

$$30 = 2 \times 15$$

Now:

$$15 = 3 \times 5$$

Therefore:

$$60 = 2 \times 2 \times 3 \times 5$$

Or:

$$60 = 2^2 \times 3 \times 5$$

13. CHINESE REMAINDER THEOREM (CRT)

The Chinese Remainder Theorem is used to solve multiple modular equations.

Example:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

We need to find a number that satisfies both equations.

Already covered in class.....